AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 10.6a: Case Study 5 — Inside the Sprint (Setup)

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Topic 10.6a Agenda

1. **Inside the Sprint — Setup —** problem framing, environment setup, early decisions

Case Study 5

Inside the sprint — anatomy of an agentic research workflow, April 15–18, 2026

What Case 5 Adds to Case 4

Case 4 showed the product

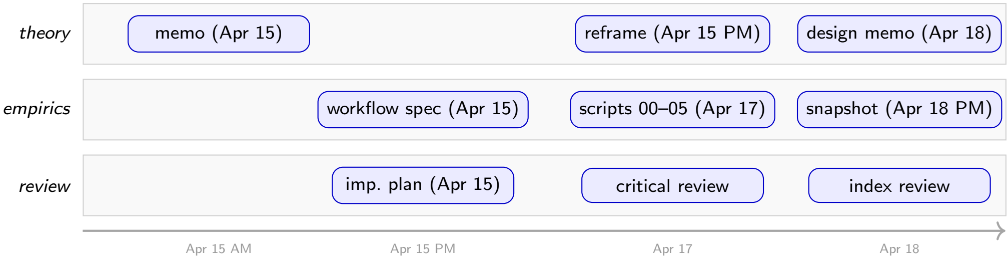
- research question (glue work + AI labor market)
- 2×2 hypothesis
- 11-script pipeline
- adversarial review and repositioning
- division of labor across the lifecycle

Case 5 shows the process

- how the first prompt is written
- how data are chosen and fetched
- how results are read and questioned
- how one index becomes four, then a distribution
- how LLMs themselves generate new data
- how each stage closes with a memo and a critique

Guiding claim. Case 4 answers "what did the collaboration produce?". Case 5 answers "how would a student replay it on a different research question?". **Four stages, eight weeks, one puzzle.** Stage 1 (Apr 15–18) is the four-day first sprint; Stages 2–4 layer on dispersion, an LLM-labeled sub/assist index, and the refined within-job distribution.

The Four-Day Timeline



Five acts across three lanes. 1. Framing → 2. Data strategy → 3. Execution → 4. Reflection and refinement → 5. Writing and iteration. Theory, empirics, and review run in parallel, not in sequence. The next 16 frames walk through these five acts inside Stage 1.

The Four-Stage Project Arc

Stage 1 (wk 1)
glue microfoundation
Scripts 01–14

glue snapshot

"proxy may be contaminated"

Stage 2 (wk 2)
within-job dispersion
Scripts 18–21

dispersion memo

"add a second AI channel"

Stage 3 (wk 3)
LLM sub/assist index
Scripts 22–29

AUI design + partial snapshot

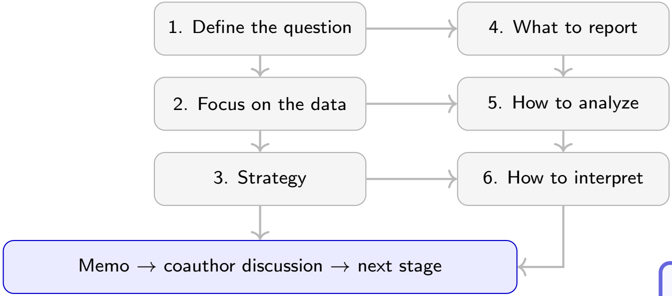
"tail is 97% correlated with $\bar{E}$"

Stage 4 (wk 4)
exposure structure
Scripts 21b, 23, 30

exposure_structure snapshot

Pattern. Each stage ends with a written memo; each transition is a *coauthor-delivered critique* of that memo. Stages are not planned upfront — each one is scoped from the previous stage's open questions. The workflow is gradual, structured, and human-corrected at every turn.

The Six-Element Stage Template



Every stage reuses the same six elements.

- **Question**: name the specific gap, not the topic.
- **Data**: name the rows and columns, not the dataset.
- **Strategy**: state the identification-*style* claim (descriptive / comparative / causal).
- **Report**: the figures and tables the stage will produce.
- **Analyze**: the spec ladder from simple to preferred.
- **Interpret**: what the result would mean.

Stage 1 (next 16 frames) is the detailed walk-through of this template. Stages 2–4 compress it into 4–8 frames each.

Act 1 — Initiating the Discussion

Opening prompt. Open with the *puzzle*, not the method. The first turn names what is unresolved in the field and invites the agent to push back on the framing before any code or measure is mentioned.

First-turn material, adapted from `glue_work_research_memo.md` §1:

> Garicano, Li, and Wu (2026) push back on task-level exposure measures by arguing that labor markets price *jobs*, not isolated tasks, so AI affects occupations through a coordination-cost channel. But their coordination cost is exogenous. This memo argues that **glue work** — the underrewarded coordination labor that preserves shared context across interdependent tasks — is the missing microfoundation for bundle strength.

What the opener buys. The agent now knows: (i) the paper is theoretical, (ii) the key primitive to be endogenized is coordination cost, (iii) the name of the candidate mechanism is glue work. Everything downstream — data choice, proxy design, figure set — is bounded by this frame.

Defining Theoretical vs. Empirical Questions

Theoretical questions

- Can coordination cost c be endogenized as a stock of coordination capital k produced by glue work?
- Does $\lambda < 1$ (partial appropriation of glue's social value) deliver a welfare externality?
- Does a second AI channel a_g acting on coordination tasks weaken bundles that a_1 alone would leave intact?

Empirical questions

- Do occupations with similar AI exposure but different glue content follow different employment trajectories after November 2022?
- Is the divergence stronger for *AI-hard* glue than for coordination in general?
- Are AI exposure and glue content empirically orthogonal enough to identify heterogeneity?

Discipline. The two question sets share vocabulary ("bundle strength", "coordination") but not evidence. Theoretical questions are answered by model writing; empirical questions by data. Never let one stand in for the other.

Why Run the Empirics First

- A 62 KB theory memo already existed on Apr 15 21:18. The theory was articulated but not stress-tested.
- Empirical illustration is **cheap to run and expensive to fake**. Running it early exposes any gap between the theory's prediction and what CPS and O*NET actually show.
- Journal readers of a theory paper with no motivating evidence do not take the framework seriously. The motivation has to be visible in the first pages.
- Case 4's headline lesson — "when data contradicts theory, reposition, don't fudge" — is only available *because* the empirics ran early enough to contradict anything.

Ordering rule. Lead with the empirics whenever the theory is already legible. The empirics act as a first round of adversarial review — cheaper than a referee and faster than a coauthor.

Sharpening — From “AI & Jobs” to One Window

“AI and the labor market”

↓

CPS-observable (occ $\times$ month)

↓

ages 22–55 in the labor force

↓

Nov 2022 pre/post cutoff

↓

FRS top quartile $\times$ glue median

Each narrowing is a decision.

- CPS-observable: rules out within-firm reallocation; only occupational movement is visible.
- 22–55: excludes student entry and retirement churn, which move for unrelated reasons.
- Nov 2022: coincident with ChatGPT’s public release; visible shock, not a causal cutoff.
- Top quartile $\times$ median: keeps Group 4 (high-AI, high-glue) large enough to support heterogeneity.

```
PRE_PERIOD_END       = "2022-10"
POST_PERIOD_START    = "2022-11"
AGE_MAIN             = (22, 55)
MIN_RAW_OBS_PREPOST  = 200
AI_HIGH_QUARTILE     = 4
```

Every narrowing is a line in `config.py`.

Act 2 — Deciding Which Data

Dataset	Role	Why chosen	Alternative (why not)
O*NET 30.2 Work Activities	glue proxy: 41 task-importance items	task-level, occupation-level, released Feb 2026	O*NET Abilities: less task-specific; cognitive labels drift faster
FRS AIOE (2021)	AI exposure score by occupation	clean SOC 2018 crosswalk; widely cited	Webb (2020): occ1990dd $\rightarrow$ Census 2018 crosswalk not yet built
IPUMS CPS monthly	employment and entry outcomes	22–55 sample, occupation codes, API access	OEWS: annual only, too aggregated to see monthly shifts
BTOS AI-use Q7	*intended* Bartik shift-share shock	sector $\times$ time AI adoption would identify exposure	Q7 suppressed at sector in all releases except 2 recent biweeks ($\sim$17%); **rejected**

Lesson. Every dataset that makes it into the pipeline has an alternative that was ruled out. Write the rejection reason down — it is the difference between a pipeline and a list of downloads.

Getting Data — API First, Manual Fallback

Automated (preferred)

- 00_download_raw_data.py fetches O*NET 30.2, the O*NET → SOC crosswalk, the BLS SOC → Census crosswalk, and the FRS AIOE appendix. Idempotent: skips files that already exist.
- 03_build_cps_occ_month_panel.py uses ipumspy and the IPUMS_API_KEY env var. It requests an extract, polls every few minutes, and downloads when ready.
- requirements.txt pins ipumspy>=0.4 so the API contract does not drift silently.
- Every fetch writes to output/logs/download_log.csv.

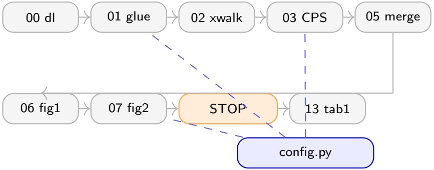
Manual (fallback)

- Webb (2020) AI exposure: MailChimp sign-up on the author's site → emailed Excel file → save to data_raw/ai_exposure/webb_ai_exposure.xlsx. Script 07 skips robustness rows if absent.
- IPUMS without API key: browse cps.ipums.org, request the same sample, drop the .dat.gz + .xml into ~/Downloads/Data-Analysis/. Script 03 autodetects.
- Failed URLs in download_log.csv name the destination, not just the URL: a human can resume where automation stopped.

The API/manual split is the one place where the agent should stop and ask the human for credentials. API keys belong in the shell profile, never in a script.

Act 3 — Running the Pipeline from Instructions

The rhythm. 51 unit tests on synthetic data *before* the first download. Each script writes a `.parquet`; later scripts read those, never the raw files. Parameters live only in `config.py`.



STOP GATE before Table 1.

```
=== Figure 2 Inspection ===
 Group 3 (High AI, Low Glue):
   pre=94.6  post=89.0  d=-5.6
 Group 4 (High AI, High Glue):
   pre=99.5  post=99.8  d=+0.2
```

- Gate passes: Groups 3 and 4 diverge after Nov 2022; Table 1 is worth computing.
- Gate fails: inspect the proxy first. A regression table before the gate is false precision.

Act 4 — When the Interaction Flips Sign

On Apr 15 22:30 the earliest `glue_core_z` run produced an uncomfortable pattern:

> Glue content **reverses sign** depending on AI exposure:
>
> - Among low-AI occupations: High Glue $\rightarrow +27$ pp better outcome than Low Glue.
> - Among high-AI occupations: High Glue $\rightarrow -15$ pp *worse* outcome than Low Glue.
>
> The interaction (AI $\times$ Glue) is negative. The simple "glue protects bundles" story predicts a positive interaction.
> — `critical_review_and_research_directions.md`

Two readings

- (a) the theory is wrong;
- (b) the proxy conflates "coordination present" with "coordination dominant" and pools AI-easy documentation with AI-hard care.

Reframe chosen

- Protection operates through occupational *reallocation*, not within-occupation survival.
- Coordination-secondary occupations (analysts) decline; coordination-primary ones (nurses) absorb the reallocation.
- The negative interaction is consistent with the theory once Condition 2 (sufficient coordination capacity) is stated explicitly.

Better Measures — the Four-Index Family

Index	What it captures	Construction	Why it matters
`glue_core_z`	baseline coordination bundle	5-item coordination average	starting diagnostic, but contaminated
`glue_aihard_z`	embodied / relational coordination	keeps only human-hard items	drops LLM-easy pseudo-glue
`glue_dominance_z`	whether coordination is central	glue items relative to full activity bundle	separates primary from secondary coordination
`glue_protection_z`	replacement-risk composite	weighted mix of hard glue, dominance, social skill, AI-easy penalty	closest summary measure for the paper

Pedagogical point. Stage 1 did not stop at one index. It built a small family of measures so the economist could ask which pattern survives once the naive glue score is decomposed.

Better Measures — AI-Hard vs. AI-Easy Coordination

AI-hard coordination items

- Communicating with supervisors, peers, and subordinates
- Establishing and maintaining relationships
- Assisting and caring for others
- Coaching and developing others
- Resolving conflicts and negotiating

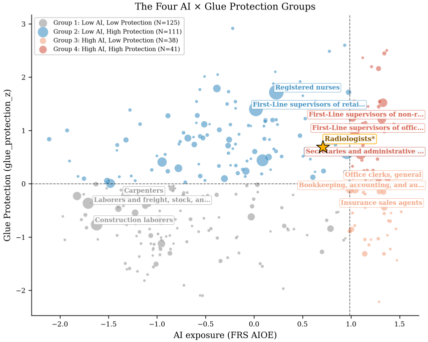
AI-easy pseudo-glue items

- Documenting and recording information
- Updating and using relevant knowledge
- Interpreting information for others

The O*NET codes still live in the project config; the slide keeps the economic distinction rather than the implementation detail.

Why split them? A radiologist and a nurse may both coordinate, but only some coordination is hard for AI to absorb. That is the distinction the refined measures are trying to preserve.

Reading Figures, Suggesting Further Analysis



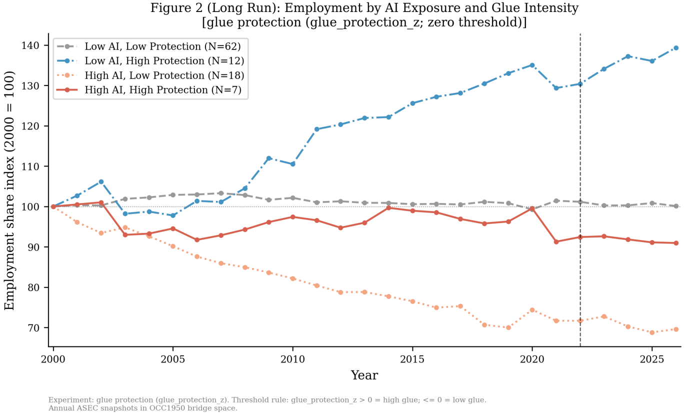
Vertical line: top-quartile AI cutoff (+0.99). Horizontal line: glue_protection_z > 0 = high glue; <= 0 = low glue (cutoff +0.00).
Bubble size ∝ pre-period employment. Each point = one Census 2018 occupation. Gold star = Radiologists, AI imputed from SOC 29-1069 ("Physicians and Surgeons, All Other").

What this quadrant plot triggers.

- AI exposure and Glue Protection are only weakly correlated.
- The high-exposure column already splits into two populations.
- Bottom-right occupations look very different from top-right occupations.
- A single high-AI average would hide the mechanism.

Next move. Compare high-protection and low-protection occupations *within* the high-AI column. That request became the differential figure.

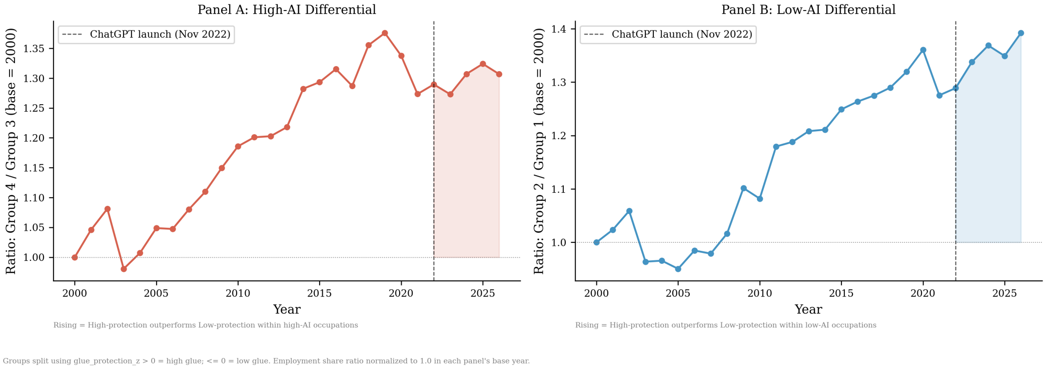
The Headline Figure — Four-Group Employment Trajectories



Reading. Once occupations are split by both AI exposure and Glue Protection, the trajectories are no longer pooled together. The theory claim is descriptive: similar AI exposure can coexist with different long-run employment paths.

The Companion Figure — Inside the High-AI Column

Figure D2: Glue Differential by AI Bucket
(Comparison of high- vs low-protection occupations within each AI bucket)



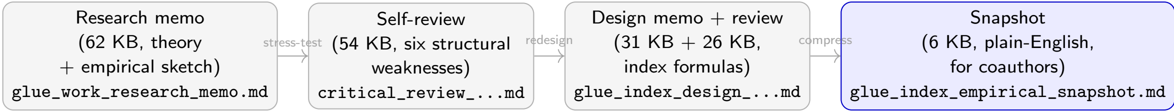
Why this matters. The ratio plot asks the cleaner question: among occupations with high AI exposure, do the high-protection occupations evolve differently from the low-protection occupations? That is the direct diagnostic behind the mechanism story.

The Diagnostic Suite

Diagnostic	What it asks	Where it lives (and what it found)
Management contamination	Is `glue_core` just measuring whether the SOC prefix is "11"?	Script 01 prints `corr(glue, is_mgmt)` and warns when > 0.4.
Sector residual (`glue_sector_z`)	Is glue just healthcare vs. production?	Partial out major-group means before re-ranking.
O*NET multi-wave (23.3 vs. 30.2)	Did task importance shift differentially in high-AI occupations?	Script 04: $\hat{\beta}_3 \approx 0$, $p \approx 0.52$. Attributed to the 2022–23 O*NET field lag.
Pre-trend decomposition (D1)	Is the divergence post-2022 or part of a long-run drift?	`figure_pretrend_allgroups_longrun_annual.png`: drift predates 2022.

Rule. Every claim in the snapshot is paired with the diagnostic that could have falsified it. The diagnostic lives on disk with a deterministic name.

Act 5 — Three-Document Writing Architecture



Each step has one job. The memo names the claim. The self-review tries to break it. The design memos rebuild the measure so the claim survives scrutiny. The snapshot delivers the survivor to a coauthor in one page. Writing one document for all four purposes produces a paper that defends everything badly.

Refining — From Memo to Snapshot

Memo opener (argumentative, for the reviewer)

> A central question in the economics of AI is whether and how artificial intelligence displaces workers. Garicano, Li, and Wu (2026) push back . . . by arguing that labor markets price *jobs*, not isolated tasks. . . . This memo argues that **glue work** is the missing microfoundation for bundle strength.

Snapshot bottom line (declarative, for the coauthor)

> Two occupations can have the same AI exposure and different labor-market trajectories. The Glue Protection index is an attempt to name one dimension along which they differ. The construction is declarative and inspectable; the next question is how to rationalize the pattern.

- Compression ratio: ~ 62 KB memo $\rightarrow \sim 6$ KB snapshot, roughly $10\times$.
- Audience shift: reviewer (argue for the framework) $\rightarrow$ coauthor (inspect the data).
- Vocabulary shift: "microfoundation" $\rightarrow$ "two occupations"; theoretical verbs $\rightarrow$ empirical verbs.

Spotting Issues, Naming Open Threads

Issue	Where surfaced	Disposition
Bartik shock infeasible	Act 2 data scan (BTOS Q7 suppressed at sector)	deferred; wait for a richer AI-adoption panel
`glue_core` contaminated by LLM-easy items	Apr 15 self-review	`glue_aihard_z` promoted to co-primary; `glue_aieasy_z` added as penalty
Coordination-presence vs. dominance	Apr 17 review	`glue_dominance_z` added (mean glue ÷ mean 41 activities)
Webb (2020) not integrated	crosswalk triage (occ1990dd → Census 2018 not built)	parked as robustness; waits on a new crosswalk
O*NET multi-wave null ($\hat{\beta}_3 \approx 0$)	Script 04 diagnostic	attributed to 2022–23 field lag; rerun on O*NET 31 when released

Each row is where the next sprint starts. A sprint that ends without a written issue list starts the next sprint by rediscovering its own unsolved problems.

Division of Labor — Move by Move

Move	Agent did	Economist did
Name the puzzle	draft the three-sentence opener from a literature sweep	chose GLW (2026) as the anchor; named glue work as the missing microfoundation
Choose O*NET items	proposed five items from the content model	vetoed three LLM-easy items; added the embodied/relational list
Build the pipeline	wrote scripts 00–14, wired `config.py`, added 51 unit tests	locked the parameter values and the STOP GATE criterion
Spot sign reversal	printed the pre/post deltas; surfaced the pattern	recognised it as a theory problem, not a bug
Reframe theory	produced three candidate rewordings	chose reallocation-not-survival; wrote the new abstract
Design the four-index family	drafted `aihard`, `dominance`, `social`, `aieasy` with weights	fixed the 35/25/25/15 weights; named the composite
Write the snapshot	compressed 62 KB $\rightarrow$ 6 KB on request	read the draft as a coauthor would; rewrote the bottom line

The constant. The agent executes; the economist decides what is true. The economist's work is not replaced — it is concentrated into judgment calls.

End of Stage 1 — Memo, Critique, and What Triggered Stage 2

The memo (6 KB)

"Two occupations can have the same AI exposure and different labor-market trajectories. Glue Protection names one possible reason. The construction is inspectable; the next step is to explain the pattern."

One page, plain English, readable by a coauthor in three minutes.

The coauthor critique

Apr 18 critique:

- "You're still using the mean of a within-occupation distribution. The *shape* matters."
- "If two tasks have very different exposures, the job gets *unbundled* — not *substituted*."
- "Add a second measure: how dispersed is the exposure *within* an occupation?"

The question that launched Stage 2

Holding mean AI exposure fixed, does within-job dispersion of exposure predict labor-share outcomes?

That question became the Stage 2 workflow over the next week.

Rule. A short memo plus an honest critique is the cheapest way to make the next sprint smarter.